DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING(A)
Osmania University, Hyderabad-07

CLASS TEST-I

SUBJECT: Data Structures

Class: B.E (CSE/AIML)

Semester: III

Date : 23-09-2025

Max Marks: 30

Duration: 1.5 Hour

S. No.	PART A Answer All Questions (10 M)	Marks	BT	CO
1. a)	Define an algorithm and list the characteristics of an algorithm.	2	3	1
b)	Differentiate linked lists and arrays	2	2	1
c)	Write the time complexity for binary search. Justify how it achieves better value than linear search	2	1	2
d)	How is circular linked list better than single linked list?	2	2	1
e)	How do you represent sparse matrices? Give an example	2	2	2
	PART B **Answer Any Two Questions (20 M)**			
2.	a) How is Asymptotic Notation useful?Compare various growth rate of functions (time complexity wise)	5	2	2
	b) Write an algorithm to sort the elements using selection sort algorithm, find the content of array after each iteration if the initial array content is 54 26 93 17 77 31 44 55 20	5	3	1
3.	a) DiscussLinear search with an example. Give the best, worst and average time complexities with explanation .	6	3	1
	b) Write a program for binary search and explain each step with an example.	4	2	1
4.	Differentiate between Singly Linked Lists and Doubly Linked Lists with suitable examples. Which is better in terms of time complexity? Explain all the operations	10	3	2

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